

Online Trajectory Optimization Method for Large Attitude Flip Vertical Landing of Starship-like Vehicles

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Abstract—An online trajectory optimization method based on convex optimization is proposed for the trajectory planning problem of large attitude flip vertical landing flight segment of a starship-like vehicle. Considering that the aerodynamic force and aerodynamic moment in the Earth landing have a large influence on the rocket flight and cannot be neglected, a planar landing motion model of a starship-like vehicle with pitching motion considering the aerodynamic influence is developed. By analyzing the constraints on its power landing process, a model of the fuel-optimal trajectory optimization problem in the landing point coordinate system is formed, and the nonconvex characteristics of the optimization model are analyzed and discussed, and a suitable sequential convexification algorithm is designed. Finally, the trajectory optimization program based on Matlab GPOPS solver is implemented for the above trajectory optimization problem, and the numerical results are analyzed and studied.

Index Terms—starship-like aircraft, large attitude flip, vertical landing, online trajectory optimization, convex optimization

I. INTRODUCTION

With the continuous expansion of human space exploration needs, providing more reliable, economical and fast spacecraft launch services is an important development direction and one of the main challenges facing the space transportation industry [1]–[3]. Launch vehicle reuse is an important technological approach to meet this challenge and has been a research hotspot among the world's major spacefaring nations [4], [5].

Since the 1960s, the United States has successively proposed a variety of ways and schemes for the reuse of launch vehicles, and carried out a large number of technical verification tests. The successful development and application of the space shuttle is a climax [6]. Since entering the 21st century, SpaceX has made a major breakthrough in vertical takeoff and landing reusable technology. Based on the continuous research and verification of many key technologies such as advanced reusable liquid rocket motors and landing guidance control, a sub-stage of its Falcon 9 rocket has taken the lead in achieving sea/land fixed-point soft landing and multiple reuse, verifying the feasibility and reliability of related technologies, and marking the stage of large-scale engineering application of VTVL reuse technology [7].

On this basis, SpaceX has proposed a fully reusable "Starship + Super Heavy" launch vehicle for larger-scale interstellar exploration missions such as Mars colonization [8]. Among them, Starship is the upper stage, which returns

and re-enters at orbital speed after completing the delivery task, causing it to face greater deceleration requirements and more complex aerodynamic and thermal environments [9]. During the high-speed phase, the glide re-entry deceleration is performed at a relatively small ballistic inclination, and this phase is controlled by multiple sets of aerodynamic rudders and RCS for composite control, followed by a soft vertical landing based on TVC control after restarting the engine and performing an attitude flip close to 90 degrees overhead near the landing point [10]. After a series of flight test exploration, in May 2021, the starship technology prototype SN15 achieved the above-mentioned large attitude flip vertical landing at the return end for the first time, verifying the relevant GNC technology and laying the foundation for the subsequent starship and super heavy combination test [11].

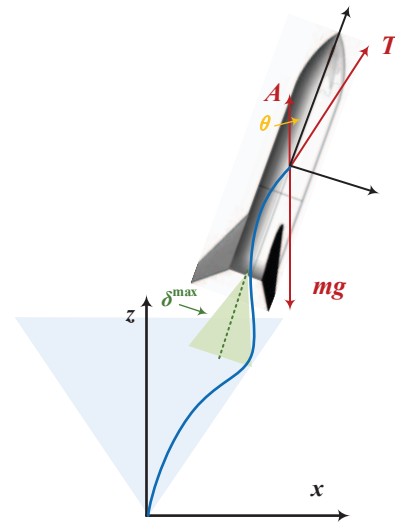


Fig. 1. Planar landing model considering pitch attitude motion.

The return-landing flight profile of the Starship and the new GNC technology it towed are of high reference value for high-speed return-vertical landing flights such as the upper stage recovery missions of reusable launch vehicles. Especially the online trajectory optimization problem of its unique "Belly Flop" large attitude flip vertical landing flight segment has high academic exploration and engineering application value. A detailed solution to this problem has not yet appeared in the

existing literature. In this paper, we investigate this problem in order to provide an onboard online trajectory optimization method for large attitude flip vertical landing of starship-like vehicles.

II. PROBLEM FORMULATION

As shown in Fig. 1, the origin of the landing point coordinate system is located at the landing point, the z -axis direction is pointed upward from the center of the earth, and the x -axis is perpendicular to the z -axis, so that the initial position and velocity vectors in the landing coordinate system lie in the plane containing the landing point. Then the equations of motion describing the planar landing of the starship-like vehicle are shown as follows.

$$\begin{aligned}\dot{x} &= V_x \\ \dot{z} &= V_z \\ \dot{V}_x &= \frac{-T \sin(\theta + \delta) + D_x}{m} \\ \dot{V}_z &= \frac{T \cos(\theta + \delta) + D_z}{m} - g_0 \\ \dot{\theta} &= w \\ \dot{w} &= \frac{(M_T + M_D)J}{T} \\ \dot{m} &= -\frac{T}{I_{sp}g_0}\end{aligned}\quad (1)$$

where $\mathbf{r} = [x, z]^T$ is the position vector, $\mathbf{V} = [V_x, V_z]^T$ is the velocity vector, θ is the pitch angle, δ is the swing angle of the engine nozzle, w is the pitch angle rate. D_x and D_z are the total aerodynamic drag in the x and z directions, respectively, described as

$$\begin{aligned}D_x &= -C_{LD} \cdot \sqrt{V_x^2 + V_z^2} \cdot V_x \\ D_z &= -C_{LD} \cdot \sqrt{V_x^2 + V_z^2} \cdot V_z\end{aligned}\quad (2)$$

where C_{LD} is the total drag coefficient. M_T and M_D are the moments generated by engine thrust and aerodynamic forces, and J is the rotational moment of inertia estimated from engineering experience about the position of the vehicle's center of mass, which is expressed as

$$\begin{aligned}M_T &= -l_{cg} \cdot T \cdot \sin \delta \\ M_D &= -(l_{cp} - l_{cg}) \cdot (D_x \cdot \cos \theta + D_z \cdot \sin \theta) \\ J &= \frac{1}{12}m(6r_s^2 + l_s^2)\end{aligned}\quad (3)$$

where r_s is the radius of the fuselage, l_s is the height of the fuselage, l_{cg} is the position of the center of mass of the aircraft, and l_{cp} is the position of the center of pressure of the aircraft. For this motion model, define its state variable as $\mathbf{x} = [x, z, V_x, V_z, \theta, w, m]^T \in \mathbb{R}^7$ and control variable as $\mathbf{u} = [T, \delta]^T \in \mathbb{R}^2$.

The state and control constraints involved in the planar landing problem of a starship-like vehicle in the landing point coordinate system are discussed below. First, consider the initial state constraint and the terminal state constraint that satisfies the fixed-point vertical soft landing as follows.

$$\mathbf{x}(t_0) = \mathbf{x}_0, \quad \mathbf{x}|_{1:6}(t_f) = \mathbf{x}_f, \quad m(t_f) \geq m_{dry} \quad (4)$$

where t_0 and t_f are the initial and terminal moments, respectively, $\mathbf{x}_0$ is the initial state navigation and sensor sampling value, and $\theta_0 = \pi/2$ is taken to characterize the

large attitude flip maneuver of the vehicle. $\mathbf{x}|_{1:6}(t_f) \triangleq [x_f, z_f, V_{xf}, V_{zf}, \theta_f, w_f]^T \in \mathbb{R}^6$, $\mathbf{x}_f$ is the terminal state, and since the landing point is the origin of the landing point coordinate system, there are generally zero rocket states except for the rocket mass which is greater than the rocket dry weight, and m_{dry} is the rocket dry weight. For the control constraints, there are thrust magnitude constraints and engine nozzle swing angle constraints as follows

$$T_{\min} \leq T \leq T_{\max}, \quad -\delta_{\max} \leq \delta \leq \delta_{\max} \quad (5)$$

Define the objective function of the optimal control problem is to minimize fuel consumption, whose performance index can be expressed as

$$J = -m(t_f) \quad (6)$$

Integrating the above mentioned starship-like vehicle motion model and constraints, the non-convex fuel-optimal trajectory optimization problem P_1 under continuous system is given as follows.

$$\begin{aligned}\min \quad & J = -m(t_f) \\ \text{s.t.} \quad & \text{Eqs. (1-5)}\end{aligned}\quad (7)$$

It can be seen that at this stage, only the motion model of the starship-like aircraft is non-convex constraints, and its initial state constraints, terminal state constraints and control constraints are all linear equality and linear inequality constraints. In the next section, we focus on non-convexity reduction model transformation methods for system dynamics.

III. TRANSFORMATION AND CONVEXIFICATION

In this section, the non-convex trajectory optimization model is initially convexified. Firstly, in order to improve the landing accuracy, make the engine nozzle swing command smoother, and uncouple the state quantity pitch angle and the control quantity engine nozzle swing angle, drawing on previous engineering experience, the engine nozzle swing rate is used as the control quantity instead of the engine nozzle swing angle, and an extended starship-like vehicle motion model is proposed.

$$\begin{aligned}\dot{x} &= V_x \\ \dot{z} &= V_z \\ \dot{V}_x &= \frac{-T \sin(\theta + \delta) + D_x}{m} \\ \dot{V}_z &= \frac{T \cos(\theta + \delta) + D_z}{m} - g_0 \\ \dot{\theta} &= w \\ \dot{w} &= \frac{(M_T + M_D)J}{T} \\ \dot{m} &= -\frac{T}{I_{sp}g_0} \\ \dot{\delta} &= d\delta\end{aligned}\quad (8)$$

For this motion model, define its state variables as $\mathbf{x} = [x, z, V_x, V_z, \theta, w, m, \delta]^T \in \mathbb{R}^8$ and control variables as $\mathbf{u} = [T, d\delta]^T \in \mathbb{R}^2$. Obtain the extended nonconvex fuel-optimal trajectory optimization problem P_2 as follows.

$$\begin{aligned}\min \quad & J = -m(t_f) \\ \text{s.t.} \quad & \text{Eqs. (2-5, 8)}\end{aligned}\quad (9)$$

where $d\delta_{\max}$ is the upper limit of the angular rate of the engine nozzle swing angle.

Then, in order to reduce the degree of nonlinearity in the extended dynamics, the component of the engine thrust in the landing coordinate system is introduced as a new control variable.

$$\begin{aligned} T_x &= T \sin(\theta + \delta) \\ T_z &= T \cos(\theta + \delta) \end{aligned} \quad (10)$$

Then the trigonometric function terms are replaced from the system dynamics. Before and after the transformation, the degrees of freedom of the system control variables are all two, but the variable substitution introduces new process constraints as follows

$$T_x^2 + T_z^2 = T^2 \quad (11)$$

$$\arctan\left(\frac{T_x}{T_z}\right) - (\theta + \delta) = 0 \quad (12)$$

In particular, (11) is naturally derived from the trigonometric relationship. The thrust component trigonometric function is determined by two parameters, namely the pitch angle θ and the engine nozzle swing angle δ , and the effect of applying the constraint (12) is to make its two relations equivalent to the original constraint form.

For the new non-convex constraint introduced by variable substitution, (11) can be handled by the lossless convexification method, that is, the equality constraint is directly relaxed into an inequality constraint

$$T_x^2 + T_z^2 \leq T^2 \quad (13)$$

Due to the time-varying mass of the starship-like vehicle during landing, the thrust-acceleration term in the velocity dynamics remains nonlinear. To convexify this nonlinear term, a new variable is further introduced to substitute it.

$$\sigma \triangleq \frac{T}{m}, T_{xm} \triangleq \frac{T_x}{m}, T_{zm} \triangleq \frac{T_z}{m}, \varpi \triangleq \ln m \quad (14)$$

Then the mass-dynamic transformation is

$$\dot{\varpi} = \alpha\sigma \quad (15)$$

where $\alpha = -1/(I_{sp}g_0)$, The mass terminal constraint transforms into

$$\varpi(t_f) \geq \ln(m_{\text{dry}}) \quad (16)$$

then there are

$$\arctan\left(\frac{T_{xm}}{T_{zm}}\right) - (\theta + \delta) = 0 \quad (17)$$

$$\sqrt{T_{xm}^2 + T_{zm}^2} \leq \sigma \quad (18)$$

$$T_{\min} \cdot e^{-\varpi} \leq \sigma \leq T_{\max} \cdot e^{-\varpi} \quad (19)$$

The system dynamics are transformed into

$$\begin{aligned} \dot{x} &= V_x \\ \dot{z} &= V_z \\ \dot{V}_x &= -T_{xm} + D_{xm} \\ \dot{V}_z &= T_{zm} + D_{zm} - g_0 \\ \dot{\theta} &= w \\ \dot{w} &= L_T \cdot \sigma \cdot \sin \delta + L_D \cdot (D_{xm} \cdot \cos \theta + D_{zm} \cdot \sin \theta) \\ \dot{\varpi} &= \alpha\sigma \\ \dot{\delta} &= d\delta \end{aligned} \quad (20)$$

where

$$\begin{aligned} D_{xm} &= e^{-\varpi} \cdot D_x \\ D_{zm} &= e^{-\varpi} \cdot D_z \\ L_T &= -12l_{cg}/(6r_s^2 + l_s^2) \\ L_D &= -12(l_{cp} - l_{cg})/(6r_s^2 + l_s^2) \end{aligned}$$

For this kinematic model, define its state variables as $\mathbf{x} = [x, z, V_x, V_z, \theta, w, \varpi, \delta]^T \in \mathbb{R}^8$, and the control variables as $\mathbf{u} = [T_{xm}, T_{zm}, \sigma, d\delta]^T \in \mathbb{R}^4$. Thus, the extended trajectory optimization problem P_3 of the non-convex fuel-optimal after lossless convexification is obtained as follows.

$$\begin{aligned} \min \quad & J = -\varpi(t_f) \\ \text{s.t.} \quad & \text{Eq.(20)} \\ & x(t_0) = x_0 \\ & x|_{1.6}(t_f) = x_f, \varpi(t_f) \geq \ln(m_{\text{dry}}) \\ & -\delta_{\max} \leq \delta \leq \delta_{\max} \\ & -d\delta_{\max} \leq d\delta \leq d\delta_{\max} \\ & \arctan(T_{xm}/T_{zm}) - (\theta + \delta) = 0 \\ & \sqrt{T_{xm}^2 + T_{zm}^2} \leq \sigma \\ & T_{\min} \cdot e^{-\varpi} \leq \sigma \leq T_{\max} \cdot e^{-\varpi} \end{aligned} \quad (21)$$

IV. NUMERICAL SIMULATION

Based on the GPOPS-II package, the trajectory optimization procedures for the above three sets of problems are prepared, and the analysis of the model characteristics and the correctness of the convexification method is carried out based on numerical experiments. The numerical simulation operating system environment in this section is Windows 10, intel i7-10710U CPU@1.8GHz, 16GB RAM.

Implementing a large attitude flip maneuver would require reducing the speed of the spacecraft by exposing the greater aerodynamic drag created by the wider surface. At about 600 m altitude, the aircraft begins a large maneuver to change its attitude from horizontal to vertical to perform a precision landing maneuver. Take the model parameters of the starship-like aircraft as follows

$$\begin{aligned} m_0 &= 120000\text{kg}, \quad m_{\text{dry}} = 85000\text{kg}, \quad l_s = 50\text{m} \\ r_s &= 4.5\text{m}, \quad l_{cg} = 20\text{m}, \quad l_{cp} = 22.5\text{m}, \quad \delta_{\max} = 20\text{deg} \\ T_{\max} &= 2210\text{kN}, \quad T_{\min} = 880\text{kN}, \quad I_{sp} = 330\text{s} \end{aligned}$$

The boundary conditions to be satisfied are as follows.

$$\begin{aligned} x(t_0) &= 100\text{m}, z(t_0) = 600\text{m}, V_x(t_0) = 0\text{m/s} \\ V_z(t_0) &= -85\text{m/s}, \theta(t_0) = \pi/2 \text{ rad}, w(t_0) = 0\text{rad/s} \\ m(t_0) &= 120000\text{kg}, x(t_f) = 0\text{m}, z(t_f) = 0\text{m} \\ V_x(t_f) &= 0\text{m/s}, V_z(t_f) = 0\text{m/s}, \theta(t_f) = 0\text{rad} \\ w(t_f) &= 0\text{rad/s}, m(t_f) \geq 85000\text{kg} \end{aligned}$$

Consider the free terminal time problem, that is, the terminal time is the optimized variable after discretization. 40 Radau collocation points were used in the preliminary optimization experiments, that is, a total of 41 discrete points. For this problem size, GPOPS-II is invoked to solve the non-convex fuel-optimal trajectory optimization problem P_1 , the augmented non-convex fuel-optimal trajectory optimization problem P_2 , and the non-convex fuel-optimal augmented trajectory optimization problem P_3 after lossless convexification for comparison experiments in the above computer

environment. The optimization results of the three problems are given below, as shown in Fig. 2-Fig. 10. By using the trajectory integration accuracy comparison method, the position accuracy and speed accuracy of the aircraft landing are shown in Table 1. Fig. 11 shows the flight trajectory of the starship aircraft flipping plane landing at a large attitude.

TABLE I
LANDING ACCURACY COMPARISON

Problem	Position Accuracy (m)	Velocity Accuracy (m/s)
P_1	2.8273	0.42003
P_2	0.42707	0.099739
P_3	0.42708	0.09974

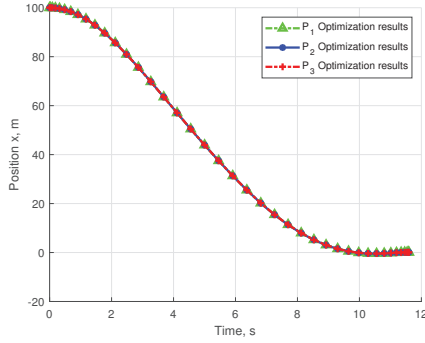


Fig. 2. Position x in the coordinate system of the landing point.

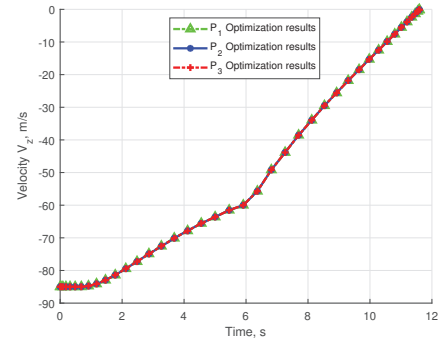


Fig. 5. Velocity V_z in the coordinate system of the landing point.

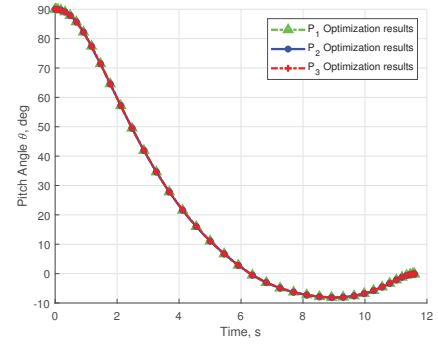


Fig. 6. Pitch angle θ in the coordinate system of the landing point.

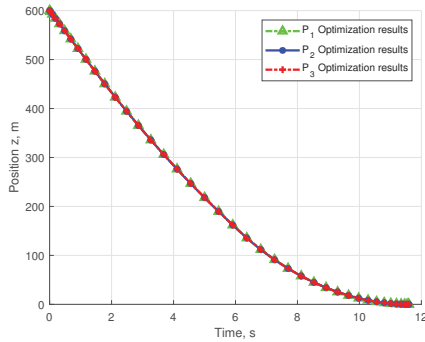


Fig. 3. Position z in the coordinate system of the landing point.

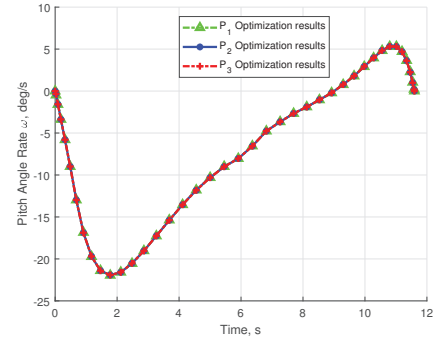


Fig. 7. Pitch angle rate w in the coordinate system of the landing point.

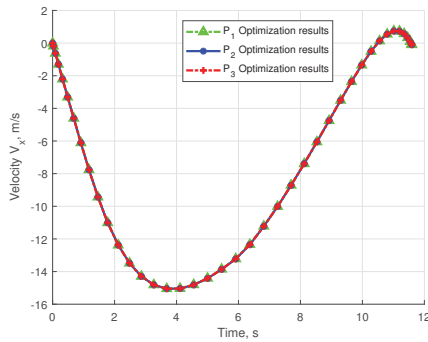


Fig. 4. Velocity V_x in the coordinate system of the landing point.

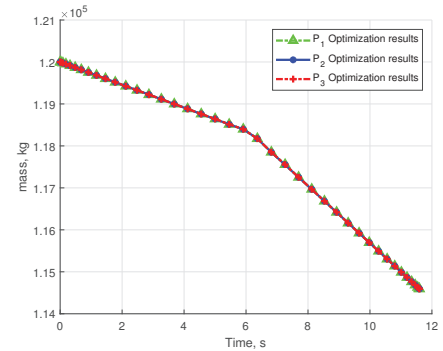


Fig. 8. Mass of the starship-like vehicle.

In summary, the preliminary analysis can be concluded as follows.

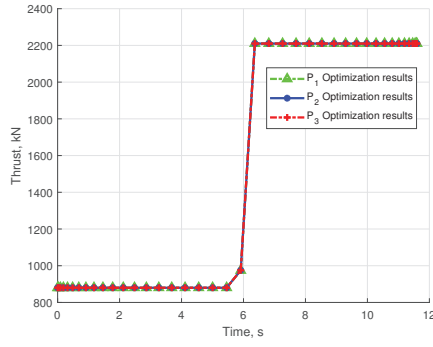


Fig. 9. Thrust of the starship-like vehicle.

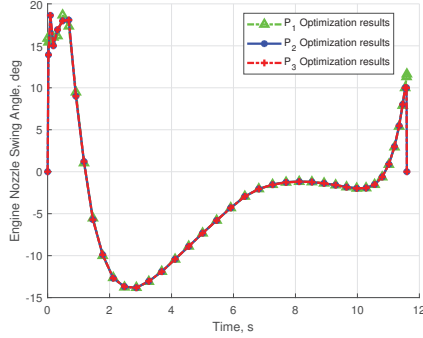


Fig. 10. The engine nozzle swing angle of the starship-like vehicle.

(1) The correctness of the current trajectory optimization model design, transformation, and partial convexification treatment is verified by invoking the GPOPS-II software package to solve the non-convex fuel-optimized trajectory optimization problem P_1 , the augmented non-convex fuel-optimized trajectory optimization problem P_2 , and the non-convex fuel-optimized augmented trajectory optimization problem P_3 after lossless convexification.

(2) The comparison of the optimization results of the P_2 problem and the P_1 problem shows that the use of the engine nozzle swing rate instead of the engine nozzle swing angle as the control quantity uncouples the state quantity pitch angle and the control quantity engine nozzle swing angle,

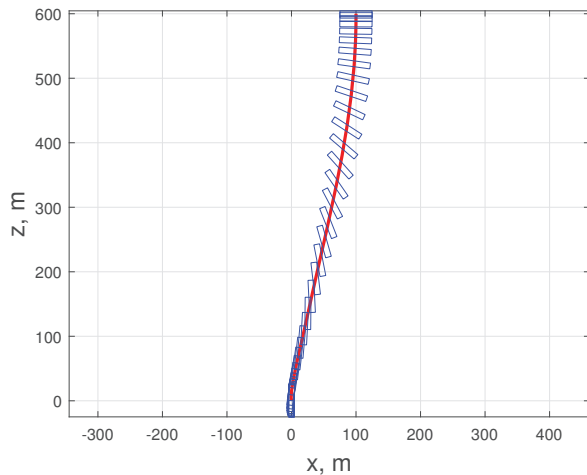


Fig. 11. Large attitude flip plane landing trajectory of a starship-like vehicle.

which makes the engine nozzle swing angle smoother and effectively improves the landing accuracy.

(3) A comparison of the optimization results of the P_3 and P_2 problems shows that the nonconvex constraint $\tan^{-1}(T_x/T_z) - (\theta + \delta) = 0$ is equivalent to the original nonconvex constraint $T_x = T \sin(\theta + \delta)$, $T_z = T \cos(\theta + \delta)$. Although it does not reduce its nonconvexity, it effectively reduces the constraint dimension without losing the additive characteristic relationship characterizing the vehicle pitch angle θ and the engine nozzle swing angle δ .

Taken together, the above conclusions indicate that the P_3 problem is the lowest non-convex problem representation before the pseudo-spectral discretization operation, and also has the conditions to form a C program by discretization and sequence convexification processing in the subsequent study.

V. CONCLUSION

Aiming at the large attitude flip vertical landing problem of the starship aircraft, this paper establishes an in-plane motion trajectory optimization model considering the pitch attitude, proposes the corresponding model transformation and partial convexification methods, and conducts numerical simulation experiments to provide the basis for the subsequent convex optimization-based optimization model. The research on on-line trajectory optimization algorithm has laid the foundation.

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